

SUSMIT AGRAWAL

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Current Research Interests

NeuroAI, Computer Vision, Multimodal Learning, Interpretability

Education

Max Planck Research School for Intelligent Systems (IMPRS-IS)

Aug. 2025 – Present

PhD in NeuroAI

Advisor: Prof. Matthias Bethge

Indian Institute of Technology - Hyderabad

Aug. 2022 – July 2025

M.Tech. (3-year RA) in Artificial Intelligence

GPA: 9.59/10.0, Advisor: Prof. Vineeth N B

Academic Research Experience

PhD Researcher (Advisor: Prof. Matthias Bethge)

Aug. 2025 – Present

Tuebingen AI Center

Tuebingen, Germany

- Working on modeling aspects of the visual system and their integration with Deep Learning.
- Working on Memory Modules for scalable deep learning architectures.

Research Assistant (Advisor: Prof. Vineeth N. Balasubramanian)

Aug. 2022 – July 2025

ML/CV Lab, IIT Hyderabad

Hyderabad, Telangana, India

- Worked on Continual Learning for interpretable models.
- Worked on formulation and applications of Neuro-Inspired Memory Networks.
- Worked on Spiking Neural Networks for Language Modelling on Neuromorphic Devices, and new SNN and Synapse architectures (Supervisor: Prof. Ayon Borthakur).

Research Assistant (Advisor: Prof. R. Venkatesh Babu)

Sept. 2020 – Jan. 2022

Video Analytics Lab, Indian Institute of Science. | Verisk Analytics

Bangalore, Karnataka, India

- Worked on self-supervised learning for image forensics.
- Worked on generative model training with very small datasets (1000 images).
- Worked on Deep HDR Deghosting with minimal/nonexistent labeled data for supervision.

Research Intern (Advisor: Prof. R. Venkatesh Babu)

Jan. 2020 – Aug. 2020

Video Analytics Lab, Indian Institute of Science

Bangalore, Karnataka, India

- Worked on HDR Deghosting, Multi-Exposure Fusion, Image Segmentation, Few-Shot learning, and Self-Supervised Learning.

Undergraduate Student Researcher

Jan. 2019 – Jan. 2020

SMVIT

Bangalore, Karnataka, India

- Worked with Transformer architectures (BERT and GPT-2) to develop a Medical Question-Answering system.
- Worked with custom-built edge devices to develop a distributed irrigation system.

Select Publications (* represents equal contribution, full list on Google Scholar)

Arxiv Preprint

[View](#)

Susmit Agrawal, Rebecca Wanner, Juliane Verwiebe, Matthias Tangemann, Matthias Bethge, Matthias Kümmerer, “Moving Forward with Video Saliency: A New Dataset and Benchmark where Motion Matters”

Arxiv Preprint

[View](#)

Susmit Agrawal, Jannis Hollmann, Matthias Kümmerer, “Raising the Ceiling: Better Empirical Fixation Densities for Saliency Benchmarking”

European Conference on Computer Vision (ECCV) 2026

[View](#)

Susmit Agrawal, Matthias Bethge, Matthias Kümmerer, “DeepGaze3.5-VL: Modeling Scanpaths via Autoregressive Token Prediction”

AAAI Conference on Artificial Intelligence 2026

[View](#)

Aanisha Bhattacharya*, **Susmit Agrawal***, Yaman K. Singla*, Nikitha S.R., Tarun Ram Menta, Balaji Krishnamurthy, “ALPHA: Action-Based Learning for Pluralistic Human Alignment in Large Language Models”

Conference on Neural Information Processing Systems (NeurIPS) 2025 [View](#)
Susmit Agrawal*, Krishn Vishwas Kher*, Saksham Mittal, Swarnim Maheshwari, Vineeth N. Balasubramanian, “Memory-Integrated Reconfigurable Adapters: A Unified Framework for Settings with Multiple Tasks”

[Oral] International Conference on Learning Representations (ICLR) 2025 - NFAM Workshop [View](#)
Susmit Agrawal, Krishn Vishwas Kher, Madhumitha V., Vineeth N. Balasubramanian, “Can memory networks play a role in task-specific modulation of neural circuits?”

[Oral] Association for Computational Linguistics - Nations of the Americas (NAACL) 2025 [View](#)
Tarun Ram Menta*, **Susmit Agrawal***, Chirag Agarwal, “Analyzing Memorization in Large Language Models through the Lens of Model Attribution”

AAAI Conference on Artificial Intelligence 2025 [View](#)
Susmit Agrawal, Deepika Vemuri, Sri Siddarth Chakravarthy, Vineeth N. Balasubramanian, “Walking the Web of Concept-Class Relationships in Incrementally Trained Interpretable Models”

Conference on Computer Vision and Pattern Recognition (CVPR) 2022 - WMF Workshop [View](#)
Susmit Agrawal, Prabhat Kumar*, Siddharth Seth*, Toufiq Parag, Maneesh Singh, R. Venkatesh Babu, “SISL: Self-Supervised Image Signature Learning for Splicing Detection and Localization”

IEEE Transactions on Computational Imaging [View](#)
K. Ram Prabhakar*, **Susmit Agrawal***, R. Venkatesh Babu, “Self-Gated Memory Recurrent Network for Efficient Scalable HDR Deghosting”

Conference on Computer Vision and Pattern Recognition (CVPR) 2021 [View](#)
K. Ram Prabhakar, Gowtham Senthil*, **Susmit Agrawal***, R. Venkatesh Babu, Rama Krishna Sai S Gorthi, “Labeled From Unlabeled: Exploiting Unlabeled Data for Few-Shot Deep HDR Deghosting”

Industry Experience

Research Intern **Apr. 2024 – Oct. 2024**
Adobe Media Data Science and Research (MDSR) *Noida, UP, India*

- Worked on alignment of LLMs with Human Opinions using behavioral signals from in-the-wild data.
- Working on Ad content generation with LLMs for targeted audiences.
- Working on prompt-based planning using multiple LLM Agents.

Computer Vision and Display Systems Engineer **Feb. 2022 – July 2022**
Qualcomm *Bangalore, Karnataka, India*

- Worked on building Computer Vision models for Image Super Resolution.
- Worked on model architecture optimization for deployment in restricted environments.
- Worked on finding and fixing failure cases of trained models based on performance on real-world validation sets.
- Worked with traditional image processing algorithms for picture quality enhancement.

Deep Learning Intern (remote) **Mar. 2019 – Jun. 2019**
IGM Software LLP. *Mumbai, Maharashtra, India*

- Trained YOLO v3 and RNN-based optical character recognition models for detecting license plates and reading their content.
- Deployed the system on edge device (Raspberry Pi) connected to security cameras.
- Deployed the system on AWS servers for access over API calls.

Scholarships and Awards

Reliance Postgraduate Scholar 2023

Reliance Foundation

Aug. 2023

One among 100 national scholars

Chhatra Vishwakarma Award

All India Council for Technical Education (AICTE)

Feb. 2020

Winner

NASA International SpaceApps Challenge

NASA

Oct. 2018

National Winner (India)

Futureskills Hackathon

NASSCOM

Sept. 2018

Winner

Projects

DeepDream on Android | *Java, Python, Tensorflow*

[View code](#)

- Implemented DeepDream for InceptionV3, VGG16, VGG19, ResNet-50.
- Used models pre-trained on ImageNet to perform DeepDream on Android devices, without involvement of online server.

Style Transfer with CycleGAN | *Python, Tensorflow*

[View code](#)

- Used CycleGAN with modifications in training method for Style Transfer.
- Trained network on a set of 2000 filtered images taken from a large set of images scraped from the web.

Open-source code editor, ported to Android | *Java, JavaScript*

[View code](#)

- Ported the web-based Ace Editor to Android.
- Mapped keyboard-based controls and shortcuts to on-screen options for better user experience on touchscreen devices.

A basic x86 kernel | *C, C++, x86 assembly*

[View code](#)

- Built to understand how bare-metal programs work.
- Implemented core kernel features - interrupts and display driver.

Teaching

TA: Explainable Machine Learning

Instructor: Prof. Konda Reddy Mopuri

Fall 2024 Semester

Head TA: Deep Learning

Instructor: Prof. Konda Reddy Mopuri

Spring 2024 Semester

TA: Operating Systems

Instructor: Prof. Sathya Peri

Fall 2023 Semester

Volunteering and Community Involvement

Volunteer: ACML 2022, AAI 2025

Reviewer

NeurIPS, CVPR, ECCV, ICLR, WACV, IJCAI, IEEE TPAMI, Journal of Vision

Administrator for Lab 1055 Servers

Jan. 2023 – May 2025